Why Are We Still Working This Hard?

It takes sixteen hours to make what a forty-hour week made in 1973. Where did the other twenty-four go?

Stella Wilson

> **Drafting note, to be removed.** Drafted by Claude for Stella Wilson, in her voice. This is the third version: half the length of the second, built on one thread (where the twenty-four hours went) and ending on what there is to hope for. The longer versions are in the repository's history. Every number is held by `checks/check_essay.py`, except that the figures for ordinary pay need a table that lives on another branch. Quotations and outside facts were read at their sources, listed at the end. One statement is from general knowledge and was not confirmed against a source: that the free time ordinary people have gained has come mostly as schooling and retirement. The section "And the other six?" leans on argument and on shared experience more than on measurement, and says so. The judgments in "Is it fair?" and "Is there any hope in this?" were made on Stella's behalf and are hers to change.

What a forty-hour week of work produced in 1973 takes sixteen hours to produce today.

That isn't a forecast. It is what the official statistics say has already happened. An hour of work in American business now makes two and a half times what it made in 1973. For every forty hours that used to be needed, twenty-four have been freed.

You have probably not noticed them. The forty-hour week was written into American law by an act of 1938, and the law still says forty. Counted for each head of population, the hours worked in American business in 2024 were 10 percent higher than in 1973. As a country, we took the whole of fifty years' progress as more output and none of it as time.



Where a forty-hour week went: the hours it takes in 2024 to make and to buy what forty hours made in 1973.

It was supposed to go differently. In 1930 the economist John Maynard Keynes wrote a short essay about the world his grandchildren would live in. He predicted that "the standard of life in progressive countries one hundred years hence will be between four and eight times as high as it is to-day." And with so much plenty, he thought, we would hardly need to work at all: "Three-hour shifts or a fifteen-hour week" would be enough. He is usually said to have been right about the wealth and wrong about the leisure. Output per person in the United States in 2024 was 7.8 times its level in 1930. But look again at the sixteen hours. He was right about both. The fifteen-hour week arrived. It just didn't arrive for anybody.

I have spent a long time trying to find out where those twenty-four hours went, with a co-author, a small economy built on paper, and more national accounts than I'd like to admit. This essay is what I found. Eighteen of the hours never reached ordinary paychecks. The other six reached them in a form that couldn't be taken as time. Both happened for the same reason, which has to do with what a wage is for. The same reason explains why people are frightened of machines that ought to be good news. And it points to a way out that is simpler, and nearer, than I expected when I started.

Two notes. The numbers are American, because the United States publishes the accounts this needs. I live in Sweden and would expect the same shape here. And I'll tell you as I go what is measured, what comes out of a model, and what I can only argue or feel. For the careful, there is a section at the very end that sorts every claim.

What is a wage for?

Three things, and we rarely notice that they are different things.

A wage is a price. It tells a business what an hour of someone's time costs, so that the hour gets used where it is worth most, and it tells people which work is wanted. A wage is good at this.

A wage is also a ticket. It is how nearly everyone gets a share of what is made. Bread reaches me because I have money, and I have money because I sold some hours. That isn't a law of nature. It is an arrangement, and it has worked so well for so long that we've stopped seeing it as one.

And a wage is a place: a shape to the day, people to be among, a sign that somebody needs you.

One thing can do three jobs only while the jobs pull the same way. For two centuries they did, because of the kind of machine we had. The machines of the industrial age were strong and stupid. They could lift, spin, stamp and haul, and they couldn't see, judge or decide. Nearly everything worth making took both kinds of ability, so every new machine needed people standing next to it, and the owners of machines had to bid for those people. The price of an hour rose with everything else, and a ticket made of that price was a good ticket.

The machines being built now can see, read, write and decide. I write software for a living and use them every day, and I find it harder each year to name a kind of work that is safely out of their reach.

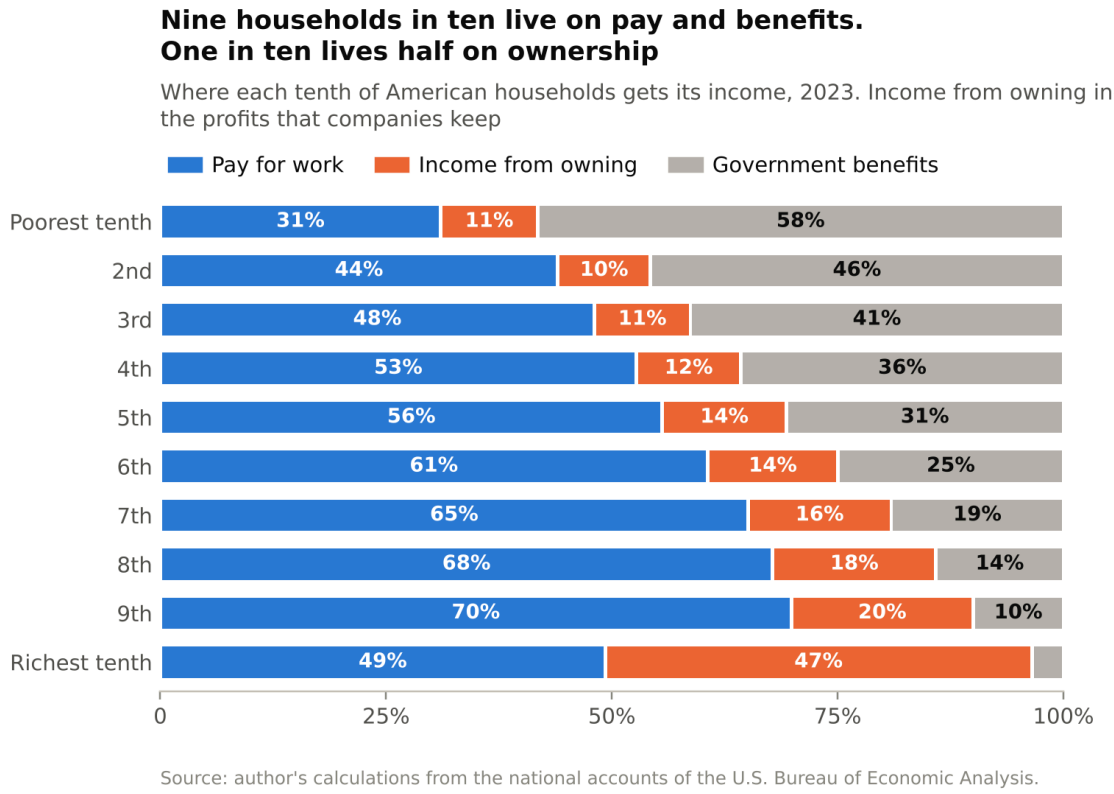
My claim is that the price and the ticket have been pulling apart for fifty years, and that each is damaging the other. The price shrinks the ticket. That is the eighteen hours. The ticket spoils the price. That is the six.

How much rides on it?

Nearly everything, for nearly everyone.

Follow a dollar you spend in a shop. Some of it pays the people who worked to make the thing. Some of it pays whoever owns what was used to make it: the oven, the truck, the building, the ground underneath. America's national accounts are detailed enough to chase a dollar all the way back through every supplier, and when you do it for 2023, of what is paid to Americans for making the things households buy, 66 percent is pay for work and 34 percent goes to owners.

A third of the economy, in other words, already pays people for owning things, with no work asked of them. That would be a curiosity if we all held a bit of both. We don't.

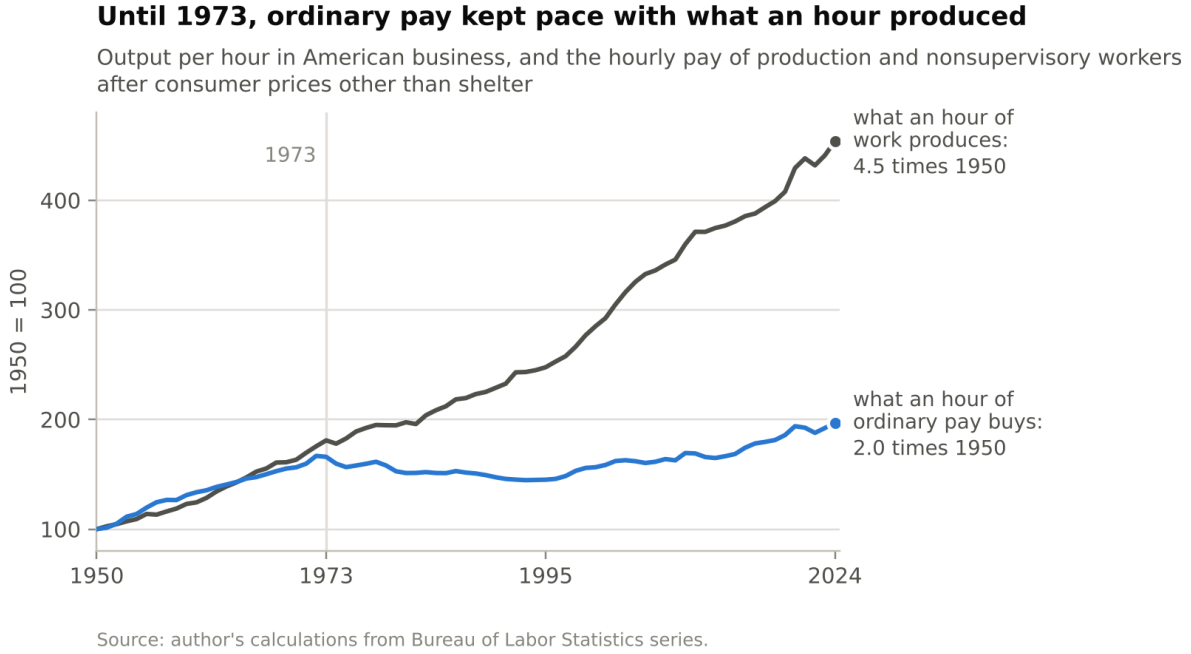


What each tenth of American households lives on, 2023.

The poorest tenth of American households lives mostly on government benefits. From the second tenth to the ninth, about four-fifths of what a household earns in the market is pay for work. Then, in the richest tenth, it changes all at once, and pay and ownership are about half and half. That one tenth receives two-thirds of all the income from ownership.

Benefits look like a second ticket for everyone else. But 68 cents of every tax dollar in America is raised from pay for work, so a benefit is mostly a claim on somebody else's wage. Follow the money far enough and nine households in ten are living on each other's wages. We live off the earth and off the machines we've built on it. Our share of what they produce reaches us through one narrow channel, which is being paid for work.

Where did the eighteen hours go?



Output per hour in American business and the hourly pay of ordinary workers, after prices, since 1950.

Until 1973, what an hour of work produced and what an hour of ordinary pay could buy rose together. Between 1950 and 1973 the first rose by 81 percent and the second by 66 percent. (By ordinary pay I mean the hourly pay of everyone on a private payroll who isn't a manager or a supervisor, measured against the prices of everything except housing.) Then they parted. Since 1973 the output of an hour has risen two and a half times, and an hour of ordinary pay buys 19 percent more.

So to buy what forty hours of ordinary pay bought in 1973 now takes thirty-four hours. Making it takes sixteen. The eighteen in between went somewhere else.

I would like to tell you they went to the owners of the machines. About a fifth of the gap did. About two-fifths of it is pay spreading out among workers, with the top pulling away from the ordinary worker. The rest I can't separate cleanly. It is a tangle of prices, benefits and measures that don't line up.

But look at what the largest piece is. Daron Acemoglu, who shared the Nobel prize in economics in 2024, and Pascual Restrepo estimate that between 50 and 70 percent of the change in America's wage structure over the four decades to 2016 came from "relative wage declines of worker groups specialized in routine tasks in industries experiencing rapid automation." Machines learned to do what ordinary workers did before they learned to do what the best-paid did.

So both pieces are the same thing happening. A wage is a price, and a price follows scarcity. It doesn't follow effort, and it doesn't follow need. The day a machine learns your job, you are working exactly as hard as you did the day before, and your hour is worth less. That is how a price ought to behave, and it is a fine way to decide who should do what. It is a strange way to decide who gets to live well.

That is the price shrinking the ticket. So far it has been slow.

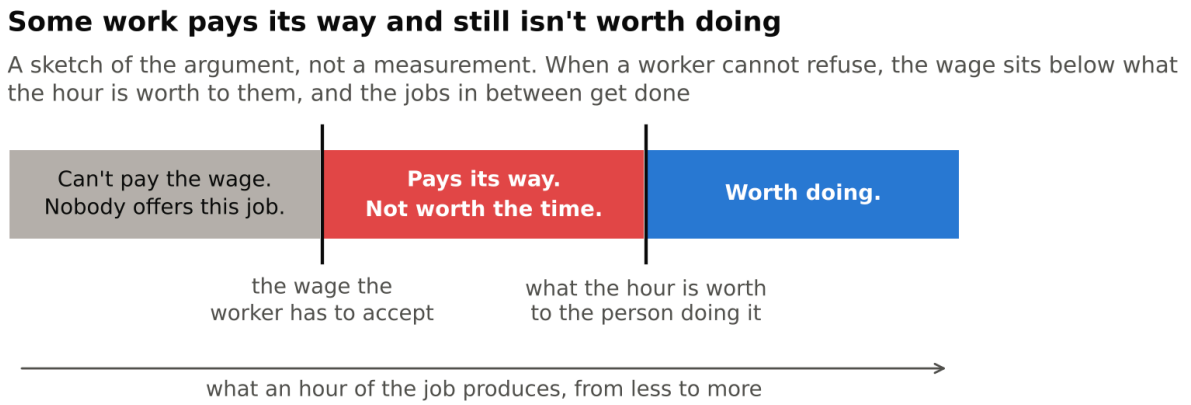
And the other six?

Ordinary pay did rise, by a fifth. That could have been six hours off the week at the same standard of living. Instead the hours went up, and I don't think it was because nobody wanted the time.

Think about what a price tells you. When I pay three dollars for a loaf of bread, the baker learns something true about me: the bread was worth at least three dollars to me, because I could have walked away. A wage is supposed to carry the same kind of information the other way. If I take a job at fifteen dollars an hour, the hour must have been worth less than that to me, or I would have kept it. A great deal of economics leans on this. It is why a job that somebody is willing to do is assumed to be a job worth doing.

It works only if I could have said no. If selling hours is the only way I eat, I will sell them for whatever they fetch, and the wage records how badly I needed the money. It says very little about what the hour was worth to me. It might have been an afternoon with my child, or a visit to my mother, or the Tuesday I would have spent running the club. Those lost a contest they were never allowed to enter. An auction in which one side can't walk away tells you very little about what the thing was worth.

This is the ticket spoiling the price. Because the wage is the ticket, most people can't refuse it, and a price that can't be refused has stopped being a price.



A sketch of the argument: jobs laid out by what an hour of them produces, against the wage and against what the hour is worth to the person doing it.

Two things follow. The first is that some work gets done that isn't worth doing. Take any job that produces a little more than the wage it pays and less than the hour is worth to the person doing it. The job covers its costs. The employer is doing nothing wrong. And every hour of it uses up something worth more than what it makes, with the whole of the loss falling on the worker. Nobody can see this, because all the prices look fine.

The second is that our economy has no way of saying "enough." Other markets can. When there is more of something than people want, its price falls and less of it gets made. The market for work can't, because for the person selling, a price that falls far enough means not eating. So we treat every job as a good in itself and every hour sold as proof that it was wanted. I once asked a friend why we assume that everyone ought to be working, and she said of course they should, she'd like to be able to buy more stuff. It's a fair answer, and it contains an assumption, which is that every extra hour worked makes more of what we want. Labor is a resource, like water, and we don't drain the oceans looking for a use for every drop.

I can't put a number on what this costs, and the reason I can't is part of the argument. Everything we measure about work is a record of what people did when they could not refuse. It is like trying to learn how much people like a food by watching what they buy when it is the only thing on the shelf. So let me ask you instead.

Do the people you know have more free time than their parents had at the same age? In a country that has made itself two and a half times as productive within one working lifetime, do they seem less pressed? When they do take time, for a new child or a dying parent, does it feel like something they are free to choose, or like something they will pay for later? And of the things that seem to have thinned out around you, the visits, the clubs, the neighbors who knew each other, how many were things that anyone was ever paid to do?

I don't offer that as evidence. I think the answers are not much in doubt, and a broken price for time predicts exactly this: that we would feel it everywhere and be unable to show it anywhere.

The harder evidence points the same way without settling the matter. Where ordinary people did gain free time over the last century, most of it came at the two ends of life, as longer schooling and longer retirement. Those are the two stretches that come with a ticket that isn't a wage: a parent or a loan at one end, a pension at the other. And when researchers in the United States gave 1,000 low-income people $1,000 a month for three years, with no conditions, those people took roughly a sixth of the money as time. They received $11,400 a year more than a comparison group, and their other income ended up about $1,900 a year lower. The authors call that "a moderate labor supply effect." I read it as a caution against my own argument as much as support for it. People who could refuse a little did refuse a little. For most of them, most of the time, more money was more urgent than more time.

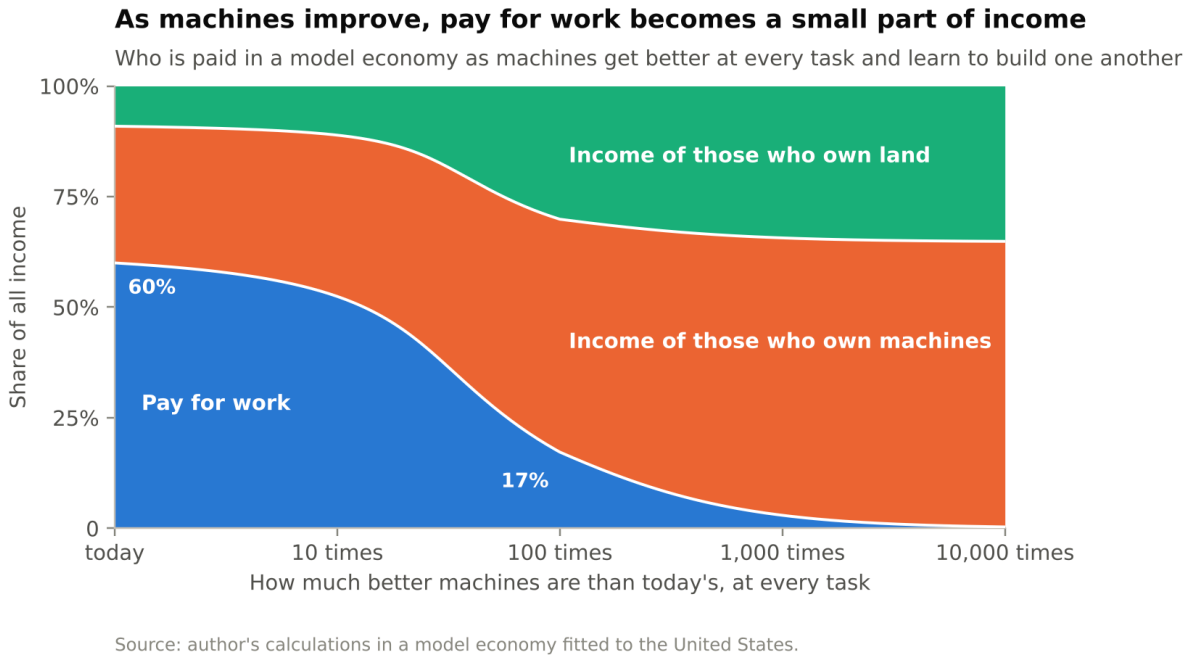
One more thing, which matters later. This is not a waste that can be recovered for nothing. The person who would gain from being able to say no has nothing to pay for the privilege with. That is their whole situation. So somebody else has to fund the floor they stand on, and a real gain arrives looking like a handout. I think that is why arguments about work and welfare go round in circles. Each side is right about the half it is looking at. Hold on to the question of who could fund that floor.

What happens if machines get much better?

So far both failures have been slow. Nobody knows whether machines are about to get much better, or how fast. What I can do is show you what happens in an economy where they do.

An economic model is a small economy on paper, simple enough to solve and fitted to the real one wherever the two can be compared. In mine, businesses give each task to whoever does it more cheaply, a person or a machine. People are far better than machines at some tasks and only a little better at others. Machines have to be built, out of other machines, some human work and some land. Whoever pays for a machine wants a return of 10 percent a year. And there is only so much land. Fitted to the United States of today, this economy pays 60 percent of its income for work, 31 percent to the owners of machines and 9 percent to the owners of land.

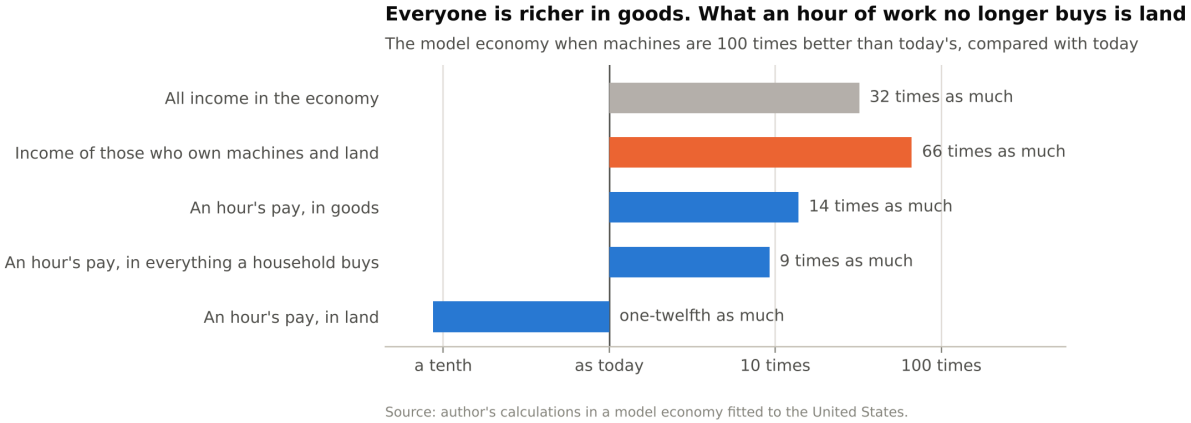
Now turn one dial. Let machines get steadily better at everything, until they are a hundred times better relative to us than they are now, and let them learn to build one another.



Who is paid in the model economy as machines improve.

For a long while very little happens. When machines are ten times better, work is still paid 52 percent of all income, because people move to what they are best at. Then, at about 35 times, the machines that could replace even the hardest hour of human work take less than an hour of human work to build, and from there no wage can stay above what the replacement costs. By the hundredfold point work is paid 17 percent of all income. At ten thousand times it is paid less than 1 percent.

Here is what I didn't expect. Nobody in this economy gets poorer.



The model economy when machines are a hundred times better, compared with today.

At the hundredfold point total income is 32 times what it was, and an hour of work buys 14 times as many goods as it does today. What an hour stops buying is land: an hour's pay rents about a twelfth of the land it does now. And the income of those who own machines and land is 66 times what it was.

Look at what has happened to the two jobs. As a price, the wage is working perfectly. It is telling everyone, accurately, that human hours are no longer what production is short of. As a ticket it has all but failed. It delivers a sixth of the income, in the middle of the largest boom there has ever been.

Who would that fall on? If ownership stayed as it is today and work were paid 17 percent of income, the richest tenth of American households would receive 62 percent of all market income, up from 46. The poorer half would go from 14 and a half percent to 10. People sometimes hope that an economy like that would seize up for want of customers. It wouldn't. The owners are the customers, and the economy would turn to face them. Nothing in economics says an economy has to be for everybody.

Keep three cautions in mind. The dial measures technology and not time: a hundredfold improvement takes 117 years at 4 percent a year, the pace at which machinery improved for its price after the Second World War, and 25 years at 20 percent, and nobody knows the pace. If people turn out to have abilities that no machine can match at any price, the same dial stops with work at 48 percent. And it is a model, with one kind of worker and no politics.

So what would fix it?

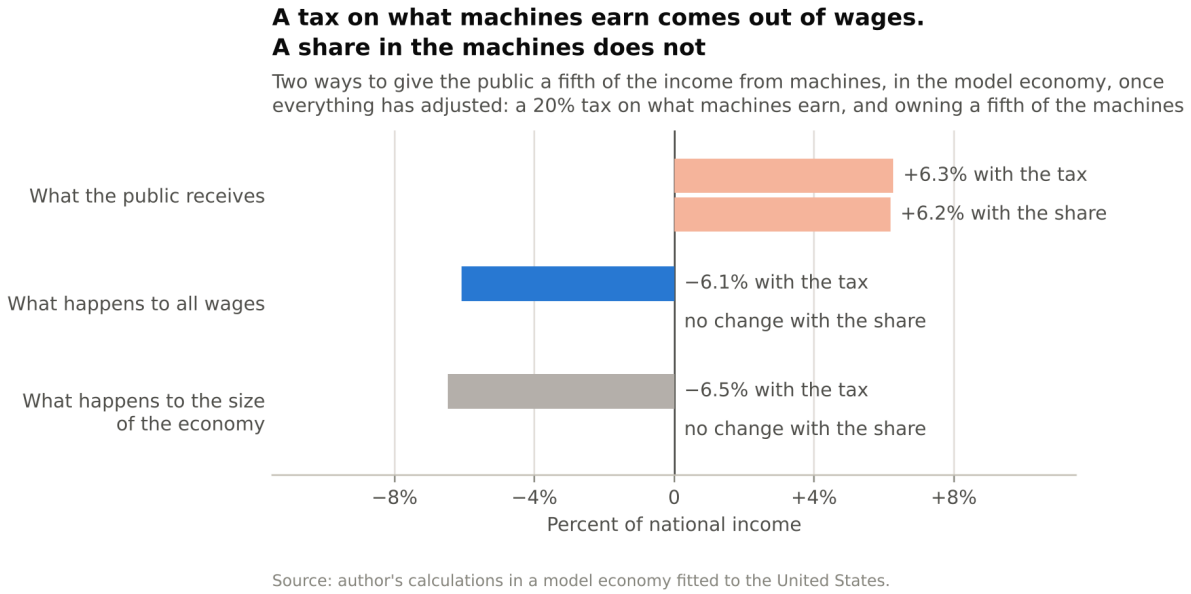
There are two problems here with one cause. We ask the wage to be a price and a ticket. Being the ticket, it can't be refused, so it has stopped pricing time. Being a price, it follows scarcity, so it is a ticket that shrinks whenever a machine learns something. The repair is to stop asking. Let the wage be a price, which it is good at, and give people a second ticket.

The question is what a second ticket can be made of.

It can't be made of wages. A benefit paid out of today's taxes is, for 68 cents in the dollar, wage earners paying each other. It moves the first ticket around.

It can't be made by taxing what machines earn, which was my own first idea. In the model, a 20 percent tax on what machines earn their owners raises 6.3 percent of national income, and wages fall by 6.1 percent of national income. The reason is that investors build machines until the last one just pays the return they want. Tax the return and they build fewer, until the ones

that remain earn the old return after tax. The owners keep their rate of return on a smaller pile. The people who work alongside the machines have fewer of them to work with, and are paid less. The tax goes straight through the owners and lands on the wage.



Two ways to give the public a fifth of the income from machines, in the model economy.

What works is owning the machines, and there is a way to do it that disturbs nothing. Let a company deduct the full cost of every investment from its taxable profit on the day it makes it. Then on every new machine the government is, in effect, putting up a fifth of the cost and collecting a fifth of what the machine earns. That is a partnership. The investor's return on their own four-fifths is untouched, so just as many machines get built. The public gets its fifth, which is 6.2 percent of national income in the model, and wages don't change. The honest catch is the machines that already exist, whose owners lose about a fifth, once.

Land is different, for one reason. Nobody makes it. Tax machines and there will be fewer machines. Tax land and there is as much land as before. Thomas Paine, the pamphleteer of the American Revolution, argued in 1797 that "Man did not make the earth," and that every landowner therefore owed the public a ground-rent, to be paid out to everyone. "It is not charity but a right," he wrote. I reached that idea on my own before I'd heard of him, which I've decided to find encouraging, and for a while I thought it was most of the answer. The numbers talked me out of it. The rent of the ground under America's homes came in 2024 to between 4.3 and 7.1 percent of households' disposable income, or between $2,776 and $4,627 a year for every person in the country. That is real money, and it isn't what a country can live on. It grows, though. In the model, land's share of income more than triples. And it can be collected by taxing only the growth from here on, which leaves today's owners the rent they have now, though not the rise they were counting on.

So the second ticket has to be made of the other third of the dollar. It has to be ownership.

What would that take?

Less than you'd think. For the United States it comes to three things.

Hold a share in common, and pay everyone its income. A fifth of all the machines and land in the country would have paid $4,493 a person in 2023, or $374 a month, for every man, woman and child. Most of that fifth is within reach without buying anything: the tax America already levies on company profits, turned into the partnership I described, and the growth in land rent together come to about a seventh of all ownership income, and a public fund could buy the rest. It isn't free money. The profits tax is already being spent, so paying it out means finding $608 billion a year somewhere else, and whatever a fund buys, somebody pays for.

Let people buy the rest out of wages. A common share of about a seventh would be enough to keep the poorer half of households whole, even if wages went to nothing. It can't do that for the comfortable middle, who earn nearly half of all the pay in the country. They have to buy their ticket, and with the common fifth in place, saving 5 percent of pay and reinvesting what it earns at 5 percent a year would do it in 34 years, which is a working life.

And move taxes off the wage, toward spending, so that the safety net stops depending on the same thing as the incomes it is there to protect.

None of this is untried. Since 1982 the state of Alaska has paid every resident a yearly dividend out of a fund built from its oil revenues. Alaska's dividend paid $1,312 a person in 2023 and $1,702 in 2024, to more than 600,000 people each year, and the fund behind it is worth $88 billion. Norway's national fund owns, in its own words, "almost 1.5 percent of all shares in the world's listed companies." And Sweden puts 2.5 of the 18.5 percent of pay that goes toward the public pension into invested accounts that belong to the worker.

It starts small. A dividend of $374 a month is 29 percent of the poverty line for a person living alone. But it is made of the thing that is growing. In the model, the income of a common fifth is 13 percent of all wages today and 96 percent of all wages at the hundredfold point. And it answers the question I left hanging. A floor paid out of something everyone owns is mostly people paying themselves. As it grows, refusing becomes possible, a little and then more, and the wage goes back to being a price that means something.

There is a deadline. Today, everything that pays ownership income in the model economy, every machine and every acre, is worth about 6.7 years of the whole country's wages. That is the price of a second ticket in the only currency most households have. At the hundredfold point it is 48

years. The saving that takes a household in the comfortable middle 34 years today takes 76 years there. The households that most need a second ticket are the ones who would have to buy it out of wages, and wages are the thing losing its value.

Is it fair?

The standard most often offered in defense of things as they are is that people are entitled to what they contribute, and that what you are paid measures what you add. I think that is a reasonable standard. My complaint is that we don't apply it.

Hold the kinds of income up to it. A wage is paid for work. The return on a machine is paid to someone who went without something in order to build it. The rent of land is paid for something that nobody made. Now look at what America taxes. It raises 68 percent of its revenue from pay for work, the most plainly earned income there is, and its tax on what machines earn lands, in the model, largely on wages again. By its own favorite standard, the country taxes most heavily what is most clearly earned. What I've described is that standard applied all the way through.

Now push it. Suppose machines can do nine-tenths of all work. The standard doesn't break. It still says that those who work should keep their pay and those who saved should keep their return. It has less and less to say, because less and less of all income is anyone's contribution. In the model with the program in place, pay for work goes from 60 percent of all income today to 17 at the hundredfold point and to less than 1 in the end. What is held in common, the public's fifth of the machines plus the growth in land rent, goes from 6 percent of all income today to 40 at the hundredfold point and 48 in the end. That is about half, and nobody chose the number. It is where the contribution standard arrives by itself when work becomes scarce.

What won't it fix?

The third job. A wage is a place as well as a price and a ticket, and a second ticket replaces none of that.

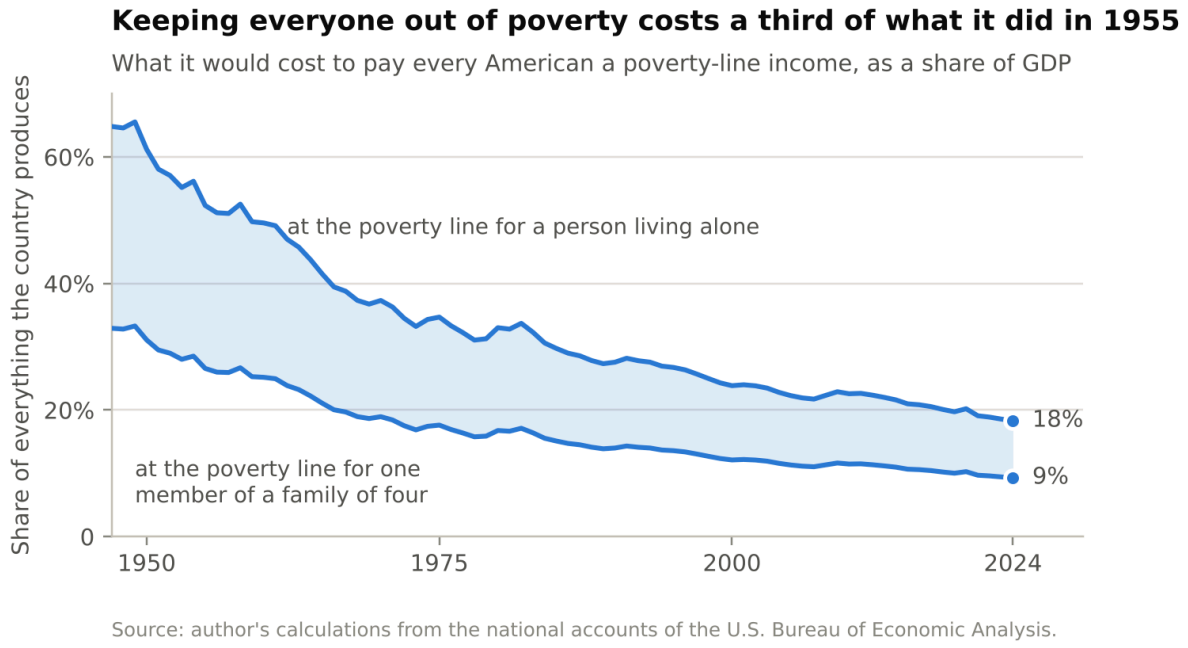
What it might do is make room. There is a room in Stockholm that I think about, a clubhouse where a gaming association lets teenagers spend their afternoons. Kids who have nowhere else to be go there. It exists because of a public grant. No business would have built it, since there is no money in a place for lonely teenagers to be less lonely. Most of what seems to have gone missing in my lifetime is like that room. None of it was ever paid. All of it takes time, and time is what a household living on wages alone has to sell.

A dividend doesn't build that room. Somebody still has to find the premises and do the boring organizing. But it lets the person who would run it on Tuesdays afford to.

Is there any hope in this?

More than I expected to find.

The first reason is the one I started with. The hard problem, the one every earlier generation faced, was making enough. That problem is very nearly solved. We make in sixteen hours what took forty within living memory.



What a poverty-line income for every American would cost, as a share of GDP, since 1947.

In 1955, paying every American a poverty-line income would have cost between 27 and 52 percent of everything the country produced. In 2024 it would cost between 9 and 18 percent. What is left is a problem of arrangement, of who holds which ticket. Arrangements are made by people, and they can be remade.

The second reason is that most of this can still be done as a trade, in which nobody is made worse off. Today, a household gets a second ticket by putting aside some of its pay and buying ownership from someone willing to sell it, and each side is paid in full for what it gives up. After a large shift, the same move is a transfer from those who gained to those who lost, which is a much harder thing to ask. We are still in the before. None of us knows for certain which side of the line our own work will fall on, and I can't think of a better condition for agreeing on a rule.

The third is that every piece of it already exists somewhere, and works.

The fourth is what the machines themselves are offering. In the small economy on paper, the same hundredfold improvement produces two different worlds, and the only difference between them is who holds the tickets. In one, a tenth of the households receive three-fifths of the income, and everyone else lives on a wage that has stopped meaning much. In the other, people own a share of what makes everything, the wage is one of two incomes, and work is something you choose. That world is 32 times richer than ours, and your time in it is your own. It is the world Keynes was describing. He assumed that a society eight times as rich would simply work less, and it turns out that working less is something you can do only if you hold something besides your work. The wealth came to us as a country. The time has to come one household at a time, and most households have had no way to receive it.

I would be glad to be wrong about where the machines are heading. Very little of this depends on that. The twenty-four hours are already there. They were freed by people who are still working forty. What stands between those people and their own time is a second ticket, and it will not get cheaper by waiting.

For the careful: how sure am I?

Measured, from public accounts, with every figure reproducible: that an hour of work makes two and a half times what it did in 1973, that an hour of ordinary pay buys a fifth more, and that Americans work more hours per head than they did then; that two-thirds of what Americans buy is paid for by work; that nine households in ten live mostly on pay and one in ten receives two-thirds of the ownership income; that 68 percent of taxes rest on pay; the size of land rent; and the falling cost of keeping everyone out of poverty.

From a model, and true to the extent that its rules hold: what happens when machines improve a hundredfold; that a tax on what machines earn lands on wages; that ownership costs more in years of wages the further the shift goes; and that the common part of income rises to about half. The model has one kind of worker, no risk and no politics. The result on taxing machines is a long-run result. In the short run the machines already exist, and their owners pay.

From argument, with modest evidence behind it: that a wage which cannot be refused says little about the worth of an hour, that some work is done which isn't worth doing, and that our economy has no way of saying "enough."

From nothing firmer than my own judgment: that machines will keep taking over tasks, and faster than they have; that a good deal of today's work would not be done if the people doing it could afford to refuse; and that most of us would take more of our wealth as time if we could.

I have left things out. Every household is given the same average shopping basket, and renters are worse placed than that makes them look. The richest tenth is lumped together, though ownership is concentrated within it. Everything here is American.

What would change my mind? The share of what we buy that is paid for by work was 71 percent in 2001 and 66 in 2023. If it rises for a decade, I am wrong about the direction. In that case what I've proposed will have cost a tax reform, a savings plan and a modest dividend, and I wouldn't regret any of them.

How this was made. I wrote this with Claude, Anthropic's AI model, which did the measurements with me, drew the figures and drafted most of the text from my earlier essays. The model economy is joint work with Johan Båge. Every number comes from code in a public repository, with checks that fail if the text and the data disagree, and every quotation was read at its source.

Sources

The measurements are my own calculations from the national accounts of the U.S. Bureau of Economic Analysis: the input–output accounts, the distribution of personal income by tenth of households, and the accounts of government receipts. Output per hour, hours worked and the wage series are from the Bureau of Labor Statistics. The poverty line is that of Mollie Orshansky, "Children of the Poor," *Social Security Bulletin* 26(7), 1963, updated for prices.

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